

The Hong Kong Polytechnic University
Subject Description Form

Subject Code	AP5006
Subject Title	Thin Film Materials and Preparation Technologies
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	This subject provides fundamental knowledge on modern syntheses of thin films. A detailed description of various thin-film deposition techniques used in industrial processing is presented. Some important industrial applications of dielectric thin-film technologies are explained.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) understand the growth parameters and limitations of thin-film deposition such as growth rate, arrival rate ratio of particles, surface energy, lattice parameters, density, stress, adhesion, stoichiometry, sticking coefficient, etc.; (b) recognize various vacuum deposition techniques to synthesize dielectric and semiconductor thin films; (c) select appropriate deposition techniques to synthesize different types of thin films with the required performance; and (d) evaluate various properties of the thin films.
Subject Synopsis/ Indicative Syllabus	Vacuum technology for thin film deposition: Working principles of different vacuum pumps such as rotary, roots, diffusion, turbomolecular, and cryogenic pumps. Concepts and methods for measuring vacuum include capacitance manometer, Pirani, and penning gauges. Design of various vacuum chambers. Deposition methods of thin-film layer: Physical vapor deposition techniques, thermal evaporation, electron beam evaporation, laser ablation, and cathode arc deposition. DC sputtering, RF sputtering, magnetron sputtering, reactive sputtering, and ion beam sputtering. Chemical vapor deposition (CVD) techniques: conventional CVD, plasma-enhanced CVD, and atomic layer deposition. Various types of thin films and their physical properties: Microstructural thin films such as single crystalline films, polycrystalline films, nano-crystalline films, amorphous films, and metastable films. Surface morphology. Film density and stoichiometry. Stress in thin films. Adhesion on substrates. Characterization of thin-film properties: Mechanical, electrical, thermal, chemical, and optical properties of thin films. Applications of thin films technologies: Transparent conducting electrodes, thin-film devices such as transistors, sensors, and solar cells, EMI shielding and RF transparent coatings for telecommunication applications, hard transparent protective coatings for optical/decorative coatings, and coatings on plastics/polymers.
Teaching/Learning Methodology	Lecture: Explain the fundamental concepts on the growth and material parameters of thin films and nanomaterials, and discuss the working principles of different thin-film techniques and their distinctive features. Tutorial: Discuss more examples to substantiate the concepts studied in the lectures,

	and help the students to understand them.					
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1. Continuous Assessments (Assignments and tests)	40%	✓	✓	✓	✓
	2. Examination	60%	✓	✓	✓	✓
	Total	100 %				
	Continuous assessments , which include assignments and tests, aims at checking the progress of students throughout the course and assisting the students in self-monitoring their performances. The examination will be used to assess the knowledge acquired by the students, as well as to determine the extent to which they have achieved the intended learning outcomes.					
Student Study Effort Expected	Class contact:					
	▪ Lecture			33 Hrs.		
	▪ Tutorial			6 Hrs		
	Other student study efforts:					
	▪ Reading and self-study			81 Hrs.		
	The total student study effort			120 Hrs.		
Reading List and References	P. M. Martin (Eds), Handbook of deposition technologies for films and coatings, Oxford, 3 rd Ed., 2009.					
	H. Lüth, Solid surfaces, interfaces, and thin films, Heidelberg, New York, Springer-Verlag, 5 th Ed., 2010.					
	K. Seshan, and D. Schepis, Handbook of Thin Film Deposition, Elsevier Science. 2018 (https://www.sciencedirect.com/book/9780128123119/handbook-of-thin-film-deposition).					